Bayesian inference - calculate probability following chains of dependencies or correlations

Bayes' Theorem (conditional probability)

$$P(A|B) = \frac{P(B|A) * P(A)}{P(B)}$$

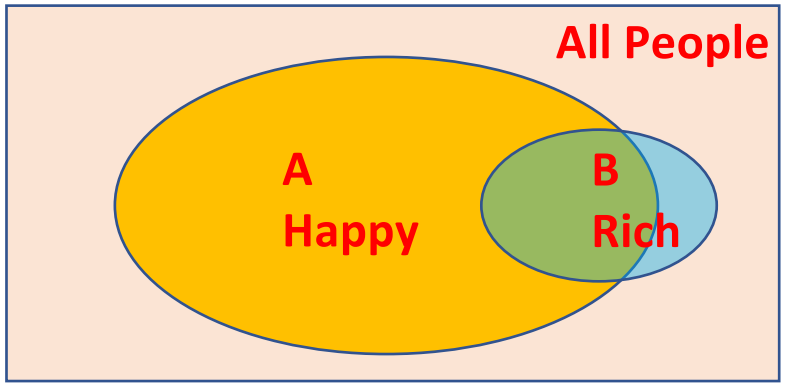
or

$$P(A|B) * P(B) = P(B|A) * P(A)$$

where A and B are events, and P(B) not equal 0.

$$\text{posterior} = \frac{\text{prior} \times \text{likelihood}}{\text{evidence}}$$

Thomas Bayes
(1701–1761)



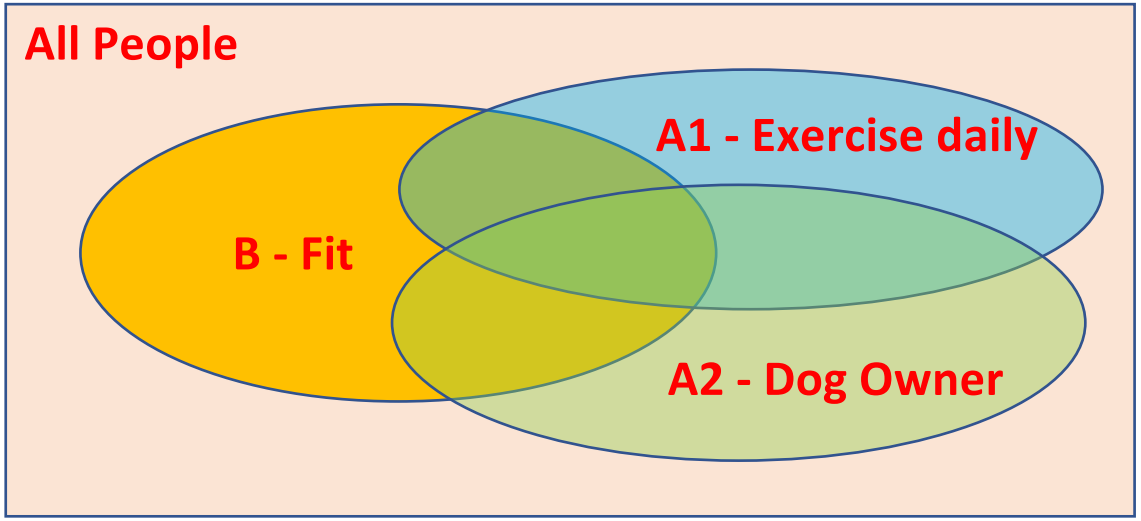
Intuitive explanation of Bayesian rule (on the left).
A = Happy, B = Rich
P(A) = 40% = probability of being happy (Yellow)
P(B) = 5% = probability of being rich (Blue)
P(A & B) = 4% = probability of rich & happy (Green)
P(B|A) = 10% = probability of being rich given that Happy
P(A|B) = 80% = probability of being happy given that rich

You can calculate green intersection using two methods:
P(A|B)*P(B) = 80%*5% = 4%
P(B|A)*P(A) = 10%*40% = 4%

Naive Bayes Classifiers

Simple probabilistic classifiers based on applying Bayes' theorem.
The main "Naive" assumption is that although individual features may have inter-dependencies and may have different importance, we "naively" assume that each feature makes an **independent** and **equal** contribution to the outcome.

On the picture below we have two features (A1 & A2) influencing outcome (B).
We will be computing conditional probability of a person being fit P(B|(A1,A2)).
We "naively" assume that A1 & A2 contribute equally to the outcome.
But in reality daily exercising probably contributes more.
Our naive model doesn't take this into account.



Bayes' Theorem

$$P(A|B) = \frac{P(B|A) * P(A)}{P(B)}$$

Features:
- A1 – Exercise Daily
- A2 – Dog owner

Outcome:
- B – Person is Fit

Naive Bayes classifiers - continued

In the table to the right we have 4 columns A_i with features and one outcome column B - "Play Golf".
For each value of the outcome (Yes or No) we can calculate the conditional probability/frequency of seeing each feature value A_i for a given outcome B:

$$P(A_i | B)$$

For example we can calculate how often we played or not played golf when it was rainy, etc.
Then we will be calculating the opposite – probability of playing golf given a specific set of features: $P(B|A_i)$. We will use Bayes formula for this.

In other words, we will be able to predict outcome B from features A_i.

A_i B

Outlook	Temp	Humidity	Windy	Play Golf
Rainy	Hot	High	False	No
Rainy	Hot	High	True	No
Overcast	Hot	High	False	Yes
Sunny	Mild	High	False	Yes
Sunny	Cool	Normal	False	Yes
Sunny	Cool	Normal	True	No
Overcast	Cool	Normal	True	Yes
Rainy	Mild	High	False	No
Rainy	Cool	Normal	False	Yes
Sunny	Mild	Normal	False	Yes
Rainy	Mild	Normal	True	Yes
Overcast	Mild	High	True	Yes
Overcast	Hot	Normal	False	Yes
Sunny	Mild	High	True	No

Here is a very good short video explaining the Naive Bayes Classification on a specific simple example:
- https://www.youtube.com/watch?v=CPqOCI0ahss
Note – there is a (common) mistake in this video! Author calculates denominators via multiplication of individual probabilities. This is wrong. You can see it by yourself how making calculation this way makes the sum of two probabilities greater than one.
Also here is a good article - https://www.geeksforgeeks.org/naive-bayes-classifiers/

Naive Bayes Classification in more details:

Suppose that an outcome "O" can have values **True** or **False** (Binary Classification) depending on events e1, e2, e3.
Suppose that influences of these events on the outcome are mutually independent.
Then we can write Bayes' Theorem formula like this:

$$P_{true} = P(True|e1, e2, e3) = \frac{P(e1, e2, e3|True) * P(True)}{P(e1, e2, e3)} = \frac{P(e1|True) * P(e2|True) * P(e3|True) * P(True)}{P(e1, e2, e3)}$$

and similarly:

$$P_{false} = P(False|e1, e2, e3) = \ldots = \frac{P(e1|False) * P(e2|False) * P(e3|False) * P(False)}{P(e1, e2, e3)}$$

Note how we write conditional probability P(e1,e2,e3|O) on top as a product of individual probabilities.
We can do that because their influences are presumed independent and equal (this is the main assumption of "Naive" approach).

Also note that the probability in the denominator is NOT split into multiplication.
We can not split it because generally we don't assume that events e1,e2,e3 are independent.
We only assume that their influences on the outcome "O" are independent (and equal in their potential influence).

But the good news is that we don't have to calculate the denominator, because it is the same for both probabilities, and for classification purposes we only need to determine which value is larger – P_{true} or P_{false}.

Also note that if you want to use this formula for only few events (for example e1 & e2), you can simply remove all other events from the formula as if you have never known that they existed. Ultimately for just one event e1 you will get classical Bayes formula:

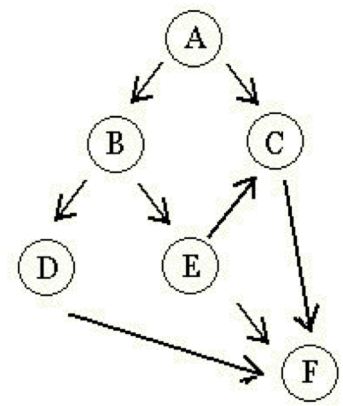
$$P(O|e1) = \frac{P(e1|O) * P(O)}{P(e1)}$$

Frequentist interpretation of Bayes' theorem.
Probability measures a proportion of outcomes.
P(A) is the proportion of outcomes with A, P(B) – with B.
P(B|A) is the proportion of outcomes with property B
out of outcomes with property A.

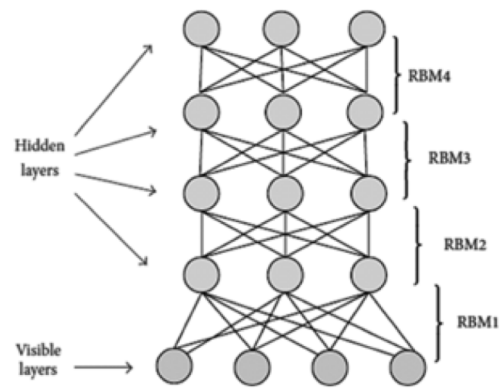
Bayesian interpretation of Bayes' theorem.
Probability measures a degree of belief.
P(A), the **prior**, is the initial degree of belief in A.
P(A|B), the **posterior**, is the degree of belief having accounted for B.
P(B|A)/P(B) – represents the support B provides for A.

Bayesian optimization
A sequential design strategy for global optimization of black-box functions that doesn't require derivatives.

Bayesian network
(Bayes network, belief network, Bayes(ian) model or probabilistic directed acyclic graphical model) is a probabilistic graphical model (a type of statistical model) that represents a set of random variables and their conditional dependencies via a directed acyclic graph (DAG).



Example:
Deep Belief Networks - made of layers of RBMs
(Restricted Boltzmann Machines),
each next layer ("B") is trained on top of the previous layer ("A").



Probability - Bayesian vs Frequentist

The **Frequentist** see probability as something that has to do with a limiting frequency based on an observed proportion.

The **Bayesian** is subjective and uses a priori beliefs to define a prior probability distribution on the possible values of the unknown parameters.

Frequentists accuse Bayesians for being subjective in selecting a prior model / distribution.

Bayesians snap back that at least they have a model, whereas Frequentists just calculate averages of observed data

Allen Downey – https://www.youtube.com/watch?v=R6d-AbkhBQ8

Frequentist (classical):

Calculate means (estimates) and confidence intervals
Problems treating missing data (**survivorship bias**)
Problems with hypothesis testing (**p-hacking**)

Bayesian:

Enumerate all possible hypotheses – and compute their probabilities.
We make a (subjective) choice of "prior" (model, distribution) and produce posterior distribution.
Allows for complex decision analysis under uncertainty.

Example1: Polling, two candidates: Alice & Bob.
52% of polled voters intend to vote for Alice.
Confidence interval [42%,62%], p-value = 0.34
What is the probability that Alice will win ?
Classical frequentist approach can not answer this question.
Bayesian method – result is a posterior distribution of possible outcomes based on data.
Thus you can get your answer.

Example2: comparing two drugs A & B.
A is better than B with p value < 0.001
A is more expensive.
Which drug should we prescribe?
Cost-benefit.
Suppose we get predicted probability of survival with A and B.
We could then compute dollars per life saved.

Example3: Guessing game
You see the prize – and guess the price
Opponent sees a different prize – and guesses the price.
Whoever is closer (without going over) wins.

Example4: MCMC (Markov-Chain Monte-Carlo)

Bayesian vs Frequentist – Survivorship Bias in World War 2

During World War II the US military's concluded that the most-hit areas of the plane needed additional armor.
But statistician Abraham Wald (Statistical Research Group (SRG) at Columbia University) has recommended adding armor to the areas that showed the least damage. His reasoning was that the sample didn't include the planes which were shot down.

It is dangerous to use data "as is" to create a model,
because the data may be biased to start with,
and it can easily lead to creating a biased model

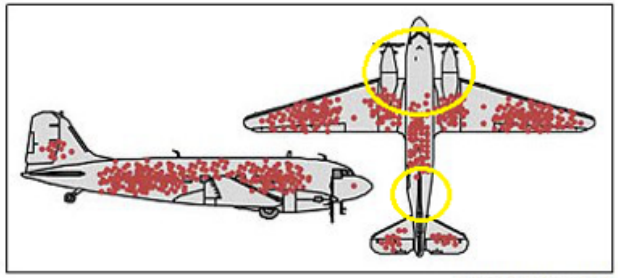
Problem occur when we have:
- under-represented features (missing data)
- over-represented features,
- "target leakage" effect.

Survivorship bias:
- https://www.youtube.com/watch?v=B3YQJ5DwTzM

Gender Bias:
- https://www.youtube.com/watch?v=p60-Jdq754A

Bias in criminal justice:
- https://www.youtube.com/watch?v=Gi4YeRqfb24

Target Leakage:
- https://www.youtube.com/watch?v=y8qaI5mpJeA



Credit: Cameron Moll

Gentlemen, you need to put more armour-plate where the holes aren't because that's where the holes were on the airplanes that didn't return - Abraham Wald 1942.

Bayesian vs Frequentist

Example: estimating θ

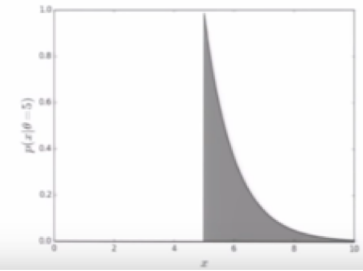
Jake VanderPlas (26 min lecture)
- https://www.youtube.com/watch?v=KhAUfqhLakw

Example - truncated exponential.
You observe times of failures of something.
The time should be less than 10 min, but for the first 10 min there are no failures because the "device" is protected. You can only measure after 10 min.
Frequentist approach just "averages" the observed values.
As all of them are >10, it gives interval 10.2 - 12.2 min
Bayesian approach with subjective flat prior gives interval 9.0 - 10.0 min, which is correct.

Uncertainties: Jaynes' Truncated Exponential

Consider a model: $p(x \mid \theta) = \begin{cases} \exp(\theta - x) & , \quad x > \theta \\ 0 & , \quad x < \theta \end{cases}$

We observe D = {10, 12, 15}
What are the 95% bounds on Θ?



Jaynes 1976

Frequentist Approach

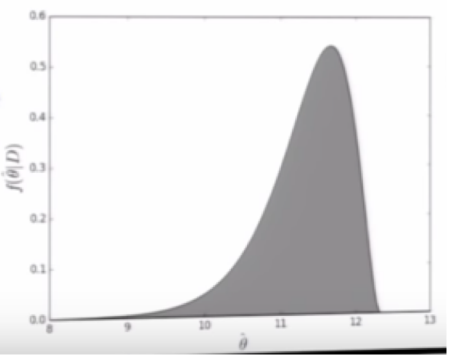
The expectation of x is:

$$E(x) = \int_0^\infty x p(x) dx = \theta + 1$$

So an unbiased estimator is:

$$\hat{\theta} = \frac{1}{N} \sum_{i=1}^{N} x_i - 1$$

Now we compute the *sampling distribution of the mean* for p(x):

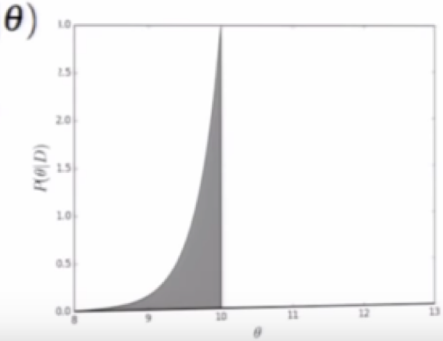


Bayesian Approach

Bayes' Theorem:

$$P(\theta|D) = \frac{P(D|\theta)P(\theta)}{P(D)}$$

Likelihood:

$$P(D|\theta) \propto \prod_{i=1}^{N} P(x_i|\theta)$$

With a flat prior, we get this posterior: